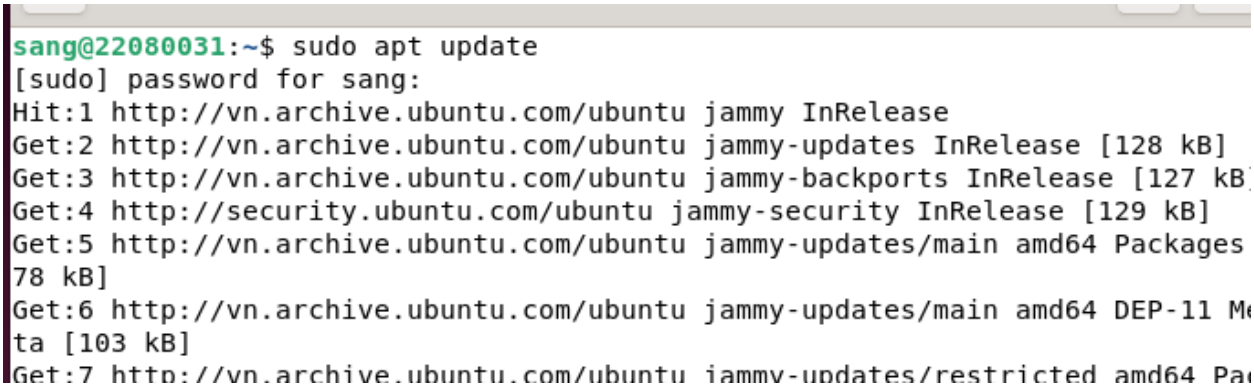
IT Infrastructure Task 3 Web Mail Directory and Switch Services

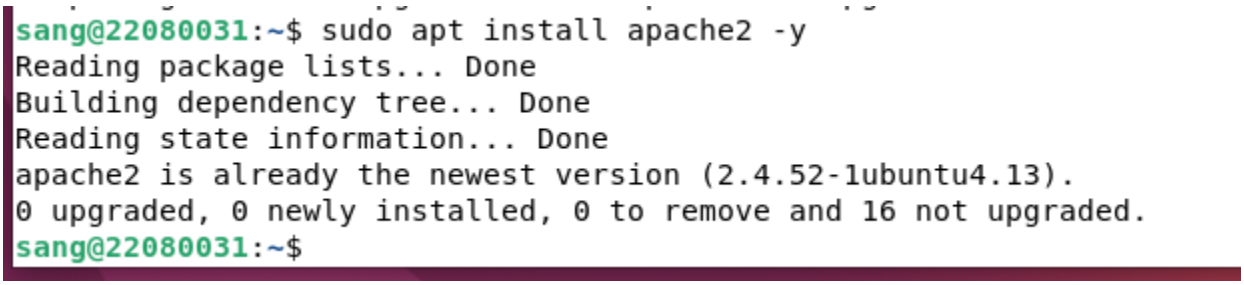
This report documents web and mail services, a Samba domain controller, and a VLAN-based switch and router lab.

Web server



The server refreshes its package lists before installing Apache. This makes apt use the configured repositories when checking the web-server package.

Task 1: install apache2 webserver



The apache2 installation command reports that Apache is already present. Apache will serve the page from the server's default document root, /var/www/html.

Start and enable the service

```
sang@22080031:~$ sudo systemctl start apache2
sang@22080031:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/system
d/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
sang@22080031:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese>
     Active: active (running) since Fri 2025-03-07 03:33:43 UTC; 7min ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1179 (apache2)
      Tasks: 55 (limit: 4555)
     Memory: 7.7M
        CPU: 97ms
     CGroup: /system.slice/apache2.service
             ├─1179 /usr/sbin/apache2 -k start
             ├─1301 /usr/sbin/apache2 -k start
```

The service is started, enabled at boot, and shown as active in systemctl. This verifies that the web process is running before a browser test is attempted.

Then allow the service to pass through firewall

```
sang@22080031:~$ sudo ufw allow "Apache Full"
Rule added
Rule added (v6)
sang@22080031:~$ sudo ufw reload
Firewall reloaded
```

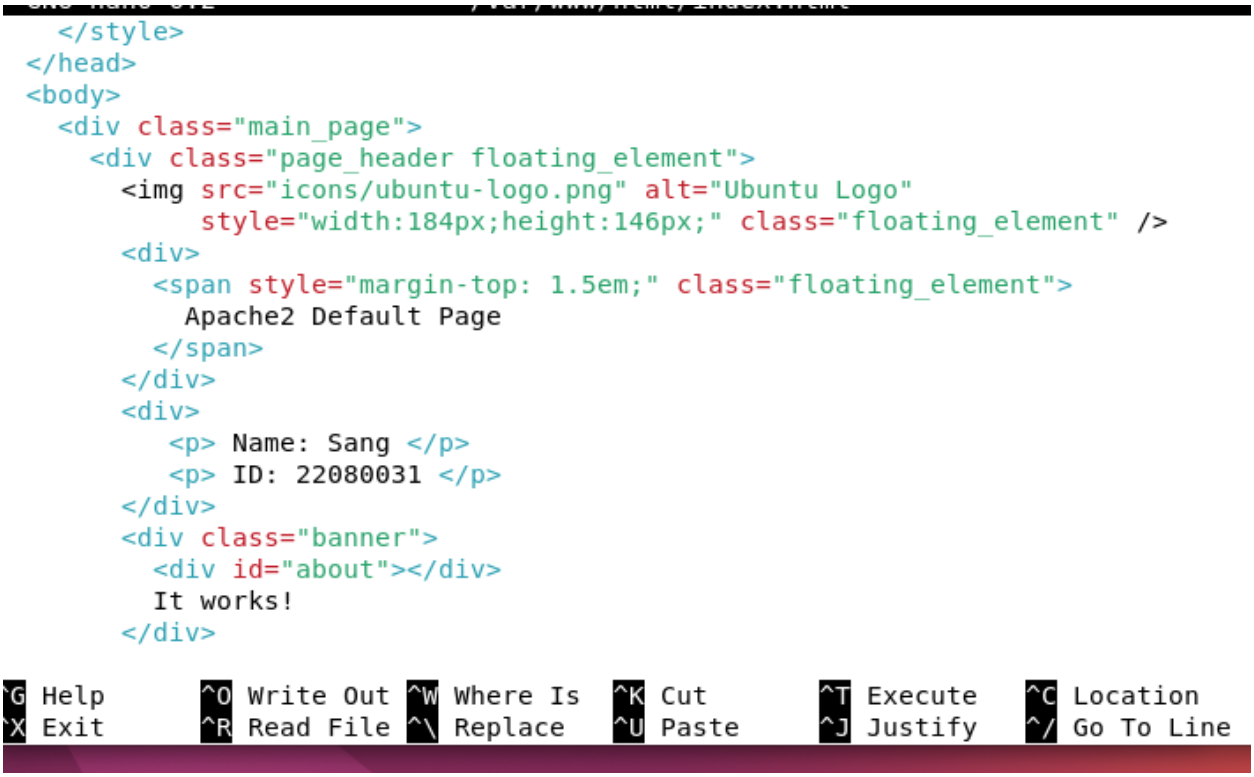
The UFW rule allows the Apache Full application profile and reloads the firewall. That profile permits web traffic to reach Apache, including the HTTP test shown later.

Task 2: Configure the index.html

Put the name and the ID number in /var/www/html/index.html

```
Firewall reloaded
sang@22080031:~$ sudo nano /var/www/html/index.html
sang@22080031:~$
```

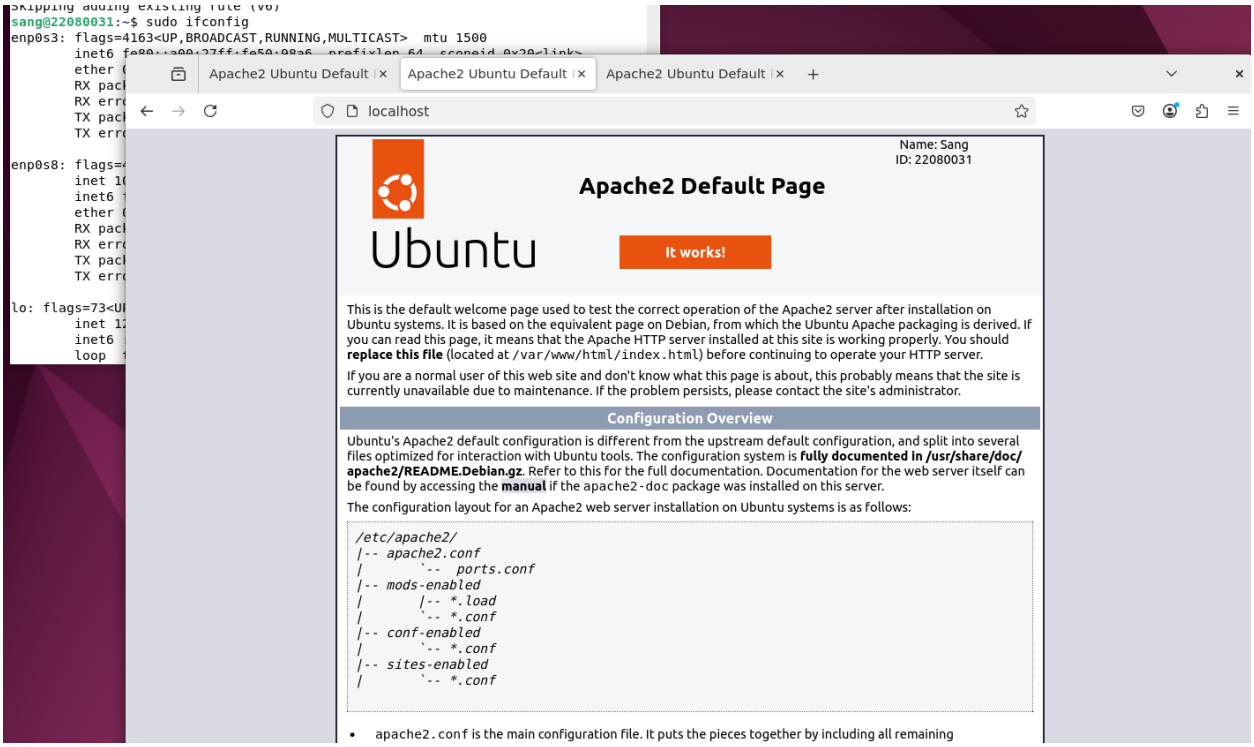
The index.html file in the default document root is opened for editing. Changing this file changes the content Apache returns for the site's home page.



The HTML capture adds the author's name and student number to the Ubuntu Apache default page. These visible fields distinguish the modified page from an untouched installation.

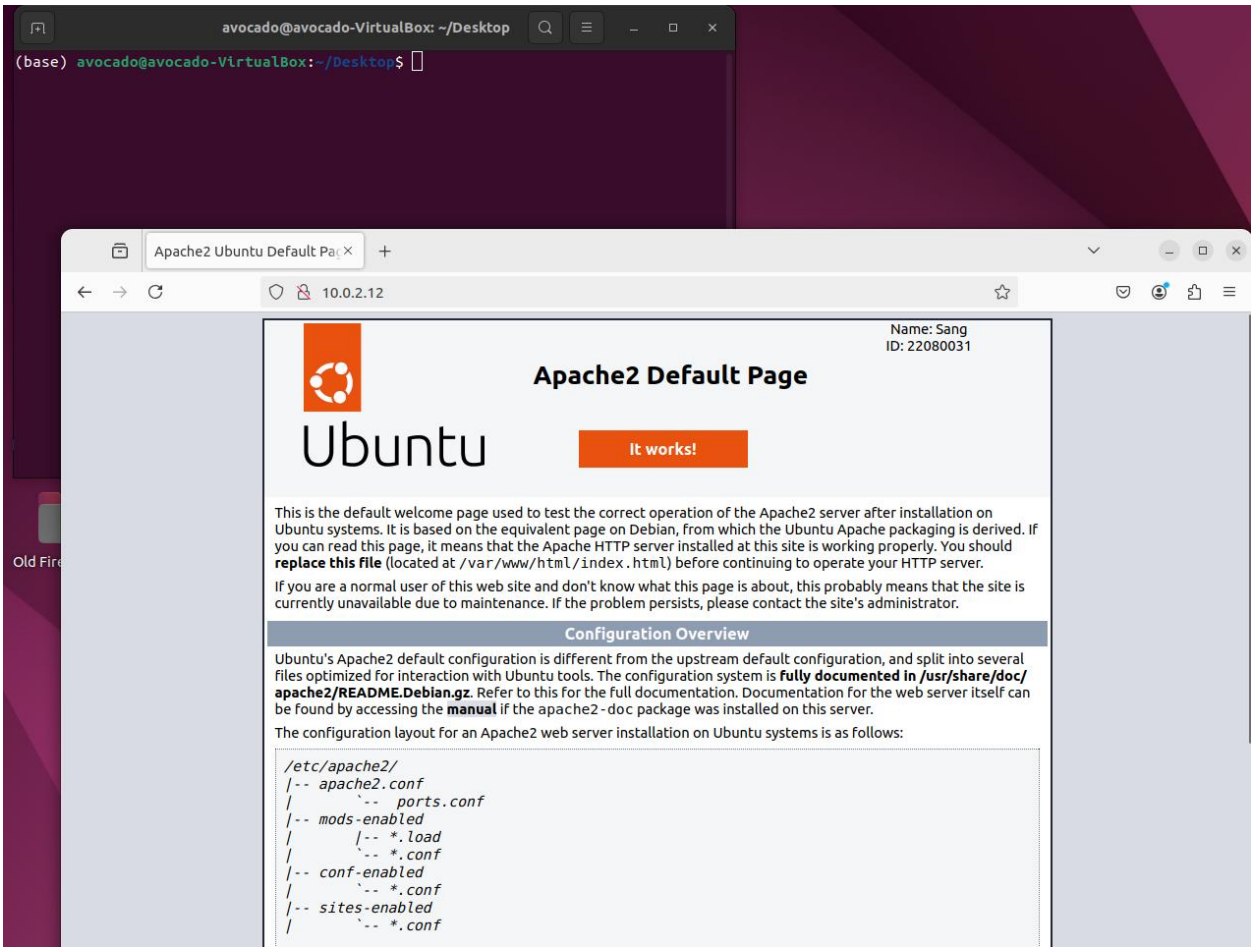
Task 3: Access the webpage

Access webpage by localhost:



The browser loads the modified page through localhost. This checks that Apache can serve the file on the server itself, but it does not yet test network access from another VM.

Access webpage by client:



The client browser loads the same page using the server's 10.0.2.12 address. Seeing the author details from the client confirms that the web service is reachable across the lab network.

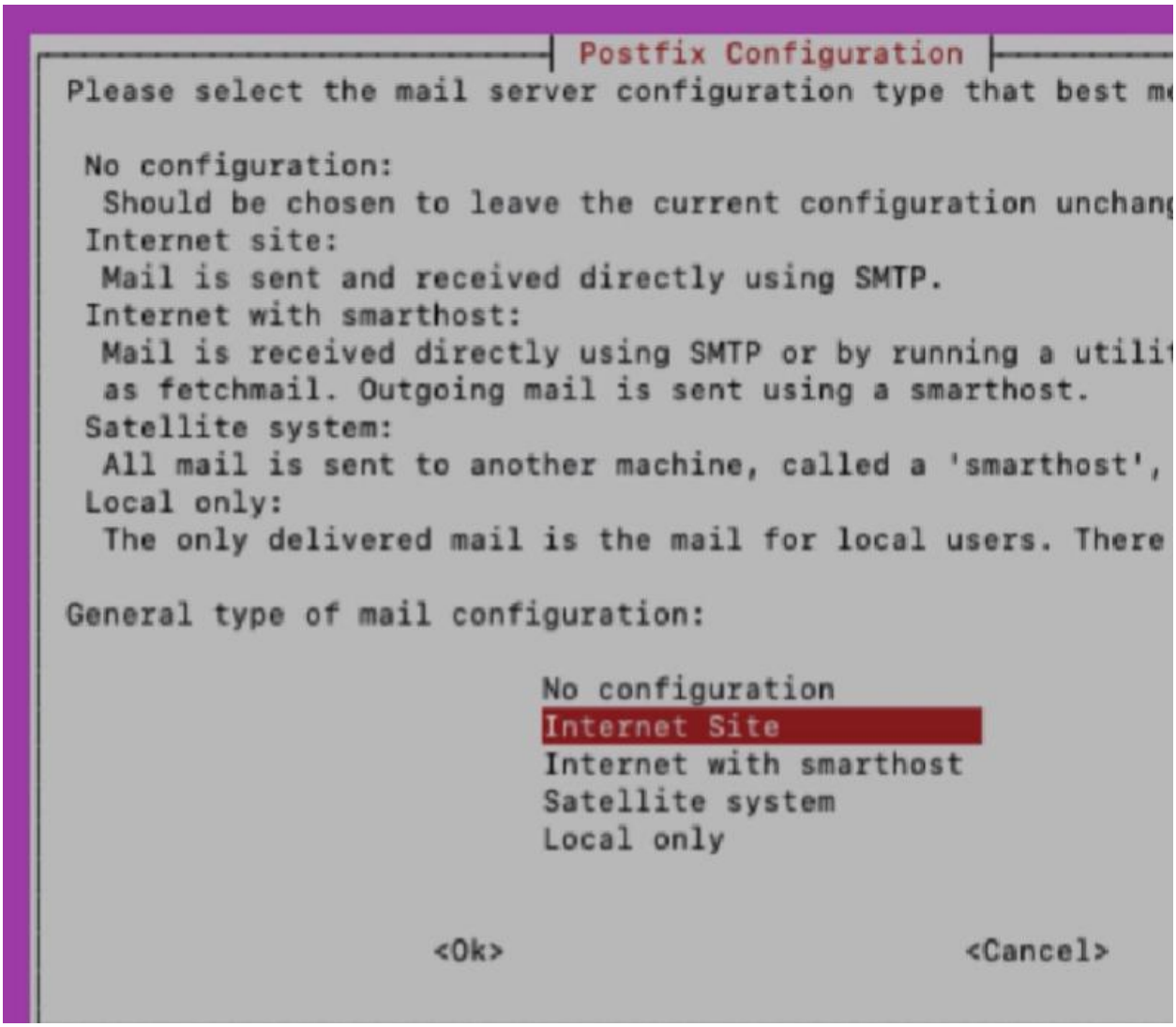
Mail Server

Task 1: Install Mail Server

```
sang@demo:~$ sudo apt-get install mailutils
[sudo] password for sang:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gsasl-common guile-3.0-libs libgsasl7 libmailutils8 libmysqlclient21
  libntlm0 libpq5 mailutils-common mysql-common postfix
Suggested packages:
  mailutils-mh mailutils-doc procmail postfix-mysql postfix-pgsql postfix-ldap
  postfix-pcre postfix-lmdb postfix-sqlite sasl2-bin | dovecot-common
  resolvconf postfix-cdb postfix-mta-sts-resolver postfix-doc
The following NEW packages will be installed:
  gsasl-common guile-3.0-libs libgsasl7 libmailutils8 libmysqlclient21
  libntlm0 libpq5 mailutils mailutils-common mysql-common postfix
0 upgraded, 11 newly installed, 0 to remove and 16 not upgraded.
Need to get 11.5 MB of archives.
After this operation, 70.4 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

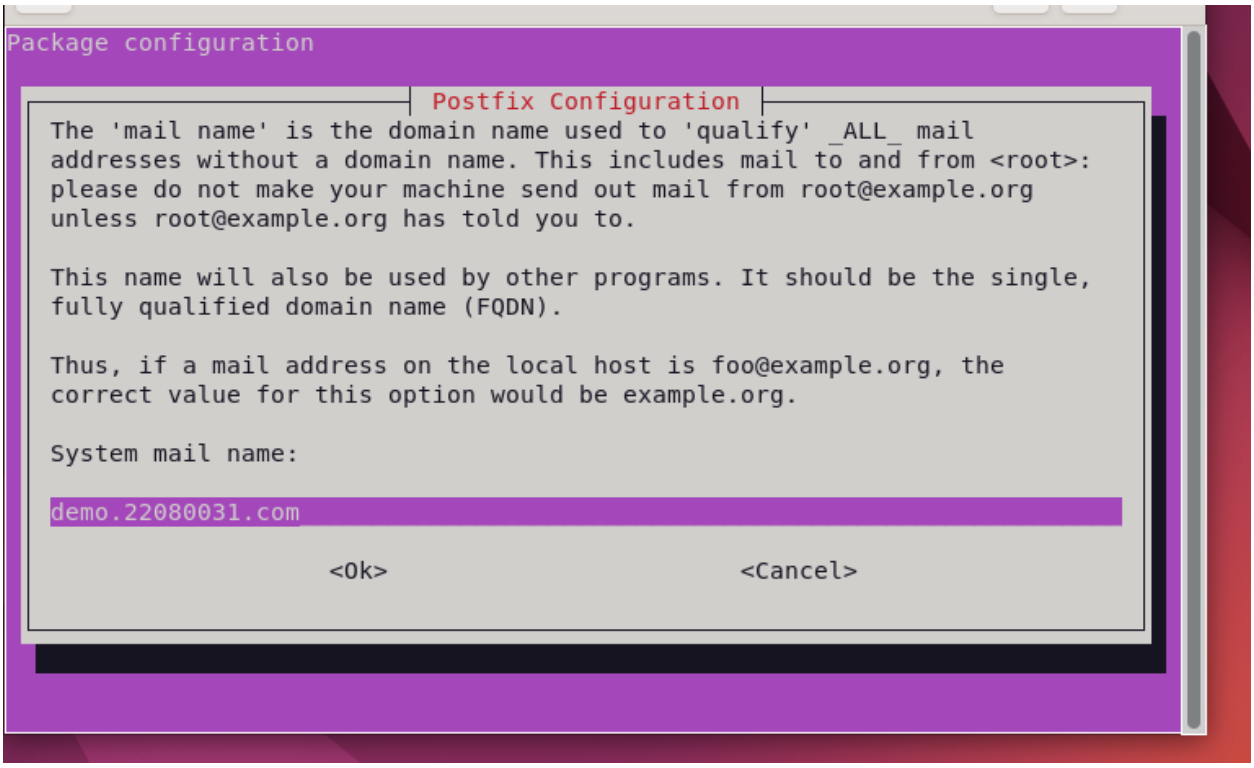
The mailutils installation also pulls in Postfix. Postfix is the mail transfer agent used for the outgoing message test.

Select ”Internet Site” to use SMTP protocol



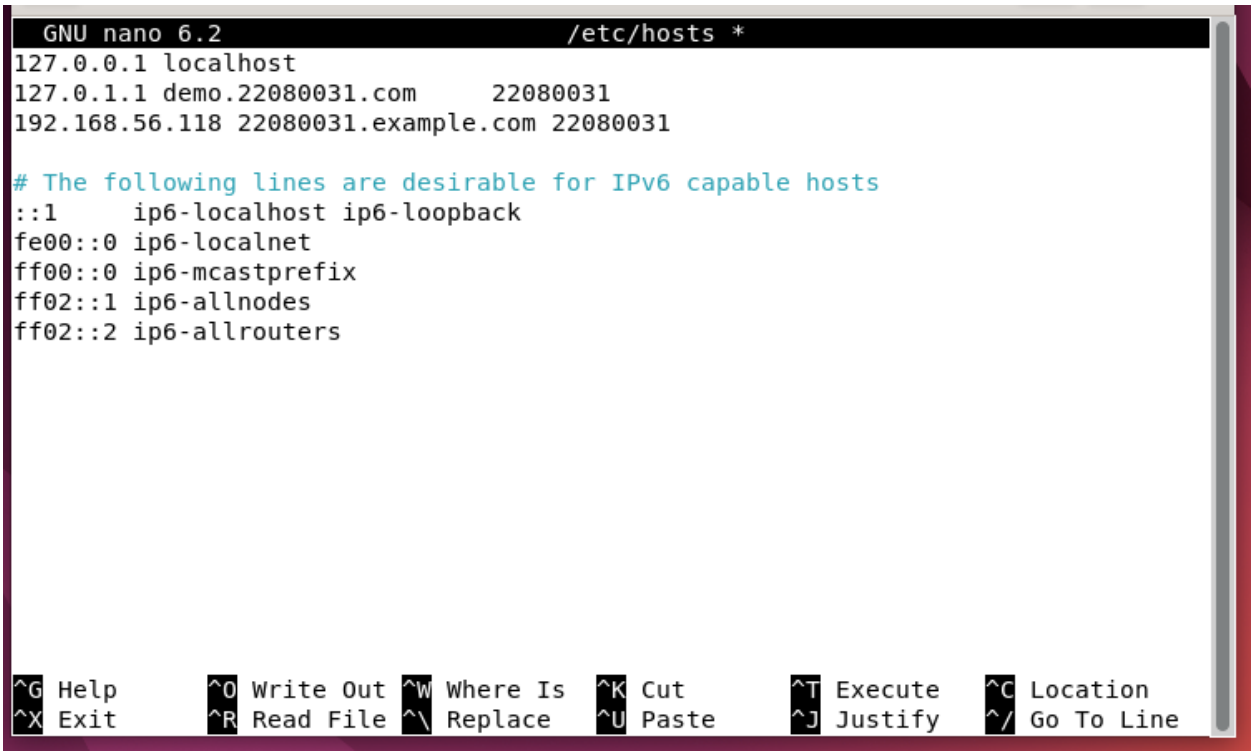
The installer selects the Internet Site mode. In this mode Postfix uses a system mail name and can handle SMTP mail rather than being limited to local mailbox delivery.

Enter the Fully Qualified Domain Name that we want to receive the email from.



The system mail name is entered as demo.22080031.com. Postfix uses this name when forming local mail addresses and identifying the mail host.

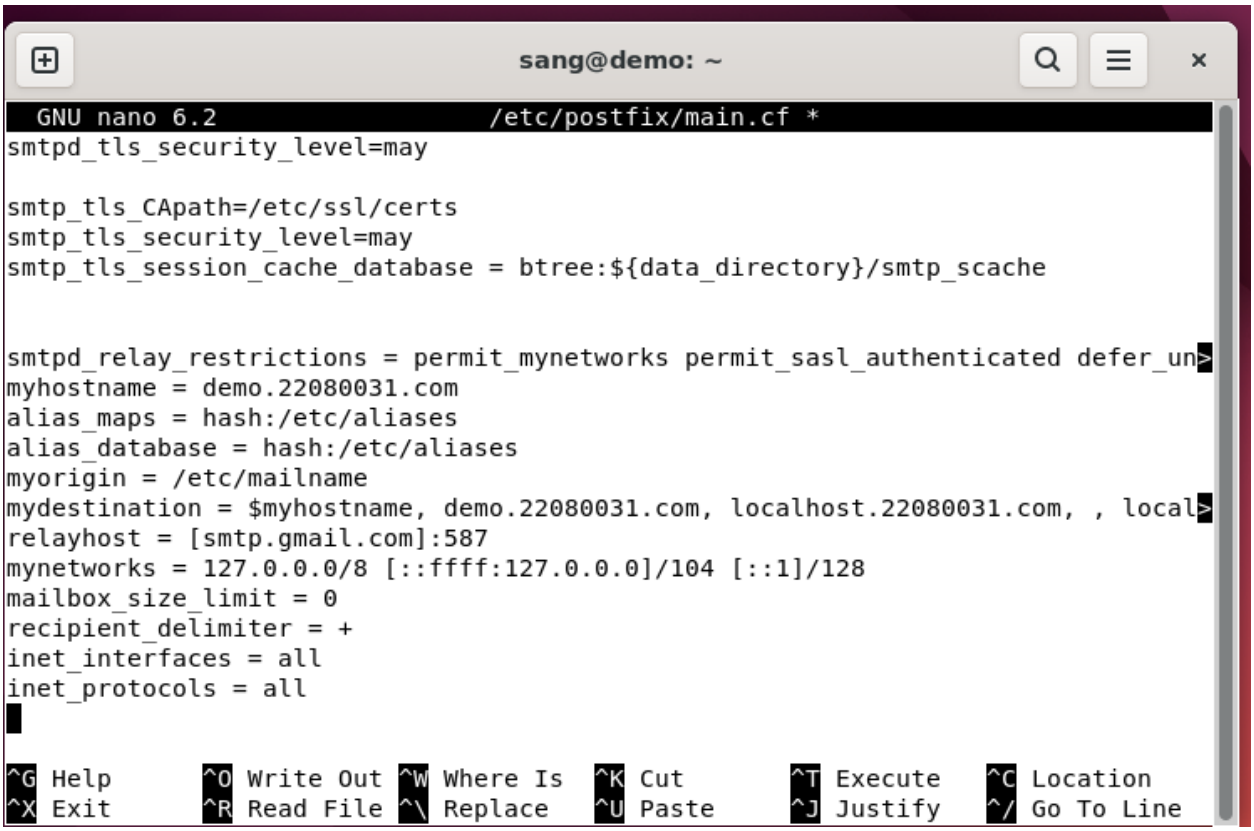
Configure the /etc/hosts file to resolve hostname (demo.22080031.com) to its IP address (127.0.1.1)



The hosts file maps demo.22080031.com to 127.0.1.1 for local resolution. This entry affects the server itself; it does not create a public DNS record for other machines.

Task 2: Configure main.cf file

In /etc/postfix/main.cf, enter the following commands:




```
smtp_sasl_auth_enable = yes
smtp_sasl_security_options = noanonymous
smtp_sasl_password_maps = hash:/etc/postfix/sasl/sasl_passwd
smtp_tls_security_level = encrypt
smtp_tls_CAfile = /etc/ssl/certs/ca-certificates.crt
```

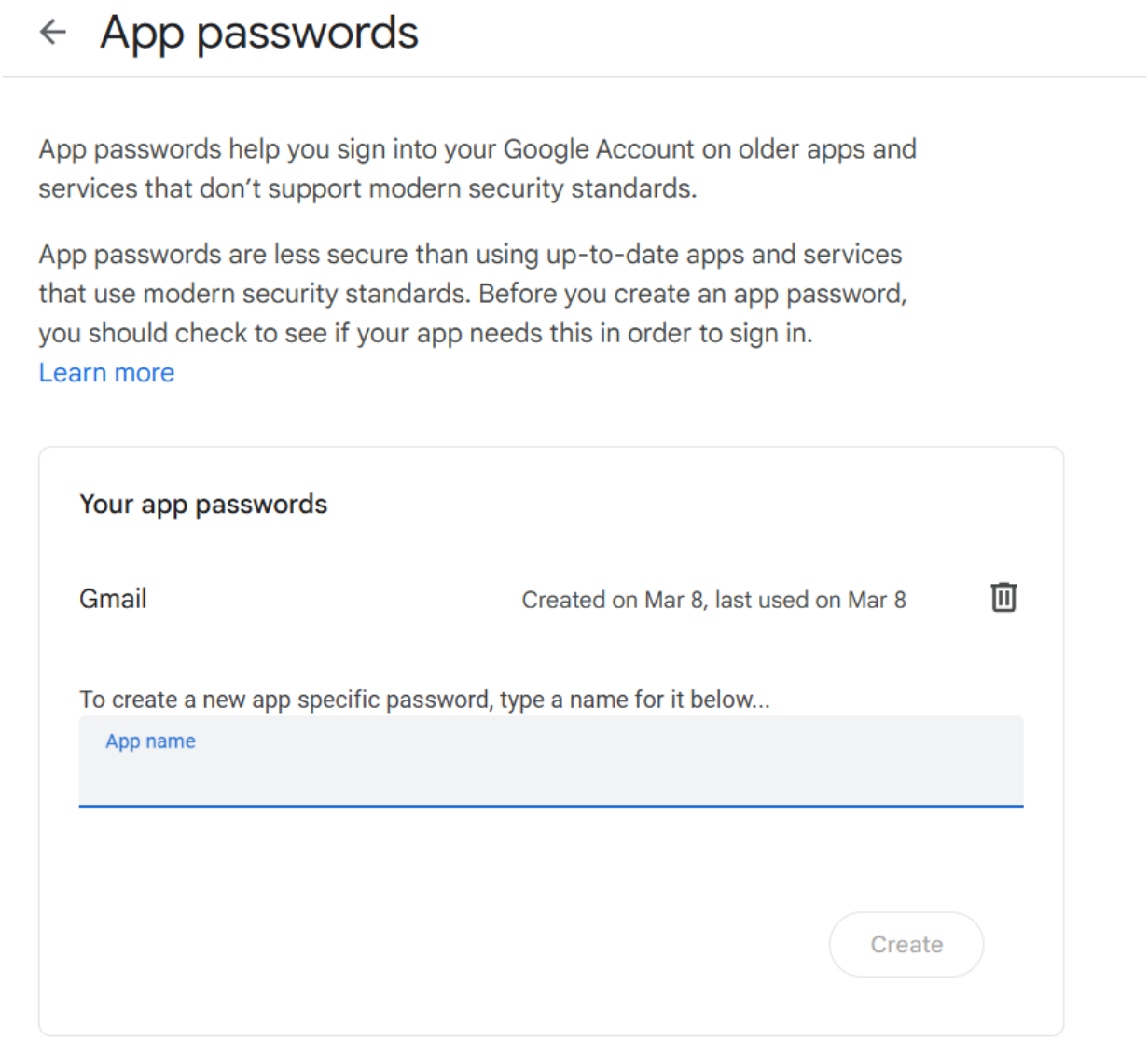
The paired main.cf captures configure the Postfix host name, accepted local destinations, and Gmail relay at port 587. They also show TLS and SASL settings used to authenticate the relay connection. The second Postfix configuration capture sets smtp_sasl_auth_enable to yes, points to the credential map, and requests encrypted transport for the relay. These options are needed for authenticated submission through the external SMTP server.

Postfix configuration

Command	Description
smtp_tls_security_level=may	Enables opportunistic TLS encryption for outgoing emails when supported by the receiving server.
smtp_tls_CApth=/etc/ssl/certs	Specifies the path to the directory containing Certificate Authority (CA) certificates for TLS verification.
smtp_tls_session_cache_database=btree:${data_directory}/smtp_scache	Configures a cache database for TLS session information to improve performance.
smtpd_relay_restrictions=permit_mynetworks permit_sasl_authenticated defer_unauth_destination	Defines rules for allowing or denying email relay based on authentication or network permissions.
myhostname=demo.22080031.com	Sets the fully qualified domain name (FQDN) of the mail server.
alias_maps=hash:/etc/aliases	Specifies the file used for email alias lookups.
alias_database=hash:/etc/aliases	Defines the alias database file that stores mail redirections.
myorigin=/etc/mailname	Defines the domain name that appears in outgoing emails from local users.
mydestination=$myhostname, demo.22080031.com, localhost.22080031.com	Specifies domains that the Postfix server will accept mail for.
relayhost= [smtp.gmail.com]:587	Configures a relay server for sending mail, using Gmail's SMTP on port 587.
mynetworks=127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128	Defines trusted networks that can send mail without authentication.
mailbox_size_limit=0	Sets no limit on mailbox size (0 means unlimited).
recipient_delimiter=+	Allows address extension using + (e.g., user+tag@example.com).
inet_interfaces=all	Enables Postfix to listen on all network interfaces.
inet_protocols=all	Supports both IPv4 and IPv6 protocols.
smtp_sasl_auth_enable=yes	Enables SMTP authentication using SASL.
smtp_sasl_security_options=noanonymous	Requires authentication but does not allow anonymous logins.
smtp_sasl_password_maps=hash:/etc/postfix/sasl/sasl_passwd	Specifies the file that contains SMTP authentication credentials.
smtp_tls_security_level=encrypt	Requires encryption for outgoing SMTP connections.

The configuration table translates the Postfix directives into their roles. The document shows smtp_tls_security_level more than once; Postfix warns that a later entry overrides an earlier one, so the final effective value should be checked.

Next go to Gmail to create app passwords then get the access passcode for relaying mail.



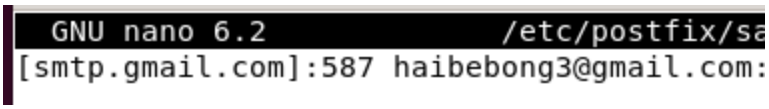
The Gmail account screen shows where an application password is created for the relay. An application password is a secret and should not be printed in a shared report or reused after exposure.

Then, go to `/etc/postfix/sasl_passwd` which is the Postfix credential store for authenticating with external SMTP servers when relaying emails.

```
sang@demo:~$ sudo nano /etc/postfix/sasl/sasl_passwd
```

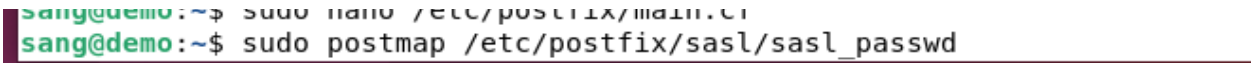
The sasl_passwd file is opened to store the relay host, account, and application password. Postfix reads this information when authenticating to Gmail's SMTP submission service.

Put the passcode with the Gmail address there:



The credential line pairs smtp.gmail.com:587 with the Gmail account and its app password. Because this is a sensitive file, its contents should be redacted before distributing the document.

Then, use postmap command to generate a hashed database file to the `/etc/postfix/sasl_passwd.db`



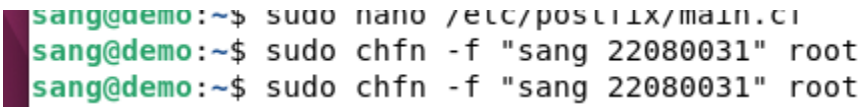
postmap creates a lookup database from the plain-text credential file. Postfix uses the generated database for the smtp_sasl_password_maps setting in main.cf.

After that, change the owner and the permission of the `/etc/postfix/sasl_passwd.db` for secure purpose. Then restart the service.



The commands change the database owner to root and restrict its mode to 600, then restart Postfix. The first attempts without sudo fail, and the later privileged commands succeed.

Then, change the email client name from root to sang 22080031:



The chfn command changes the display name associated with the local sender account. It affects the sender label seen in mail, not the relay credentials or SMTP delivery path.

Task 3: Send an email

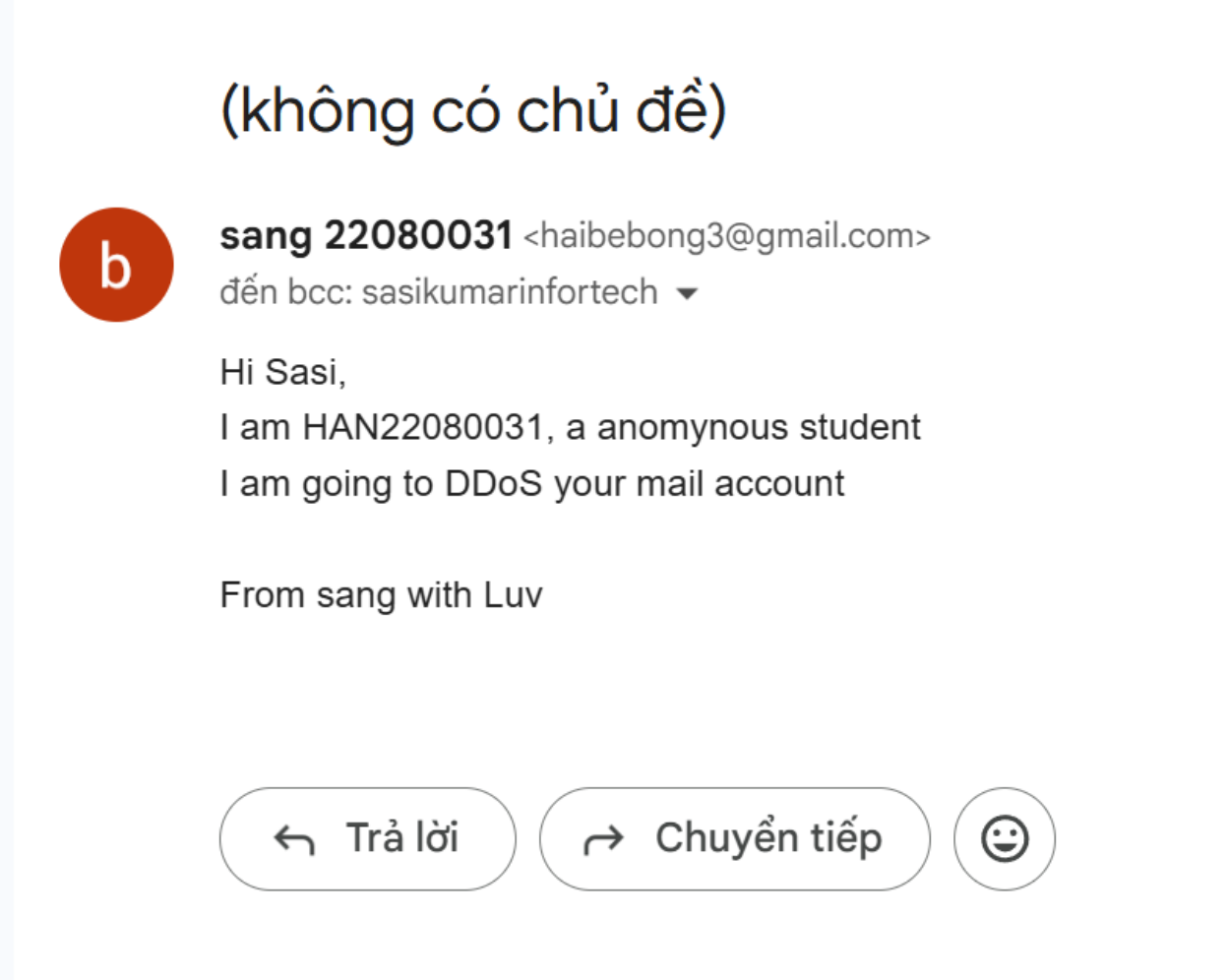
write an email to sasikumarinfortech@gmail.com

```
sang@22080031:~$ sudo sendmail sasikumarinfortech@gmail.com
sendmail: warning: /etc/postfix/main.cf, line 53: overriding earlier entry: smtp
_tls_security_level=may
postdrop: warning: /etc/postfix/main.cf, line 53: overriding earlier entry: smtp
_tls_security_level=may
Hi Sasi,
I am HAN22080031, a anomynous student
I am going to DDoS your mail account

From sang with Luv
sang@22080031:~$
```

The sendmail command submits a test message to an external recipient. The terminal also prints warnings about the repeated TLS setting; submission alone does not prove the message arrived.

sasikumarinfortech@gmail.com has received the email.



The Gmail inbox capture shows the test message delivered from the configured account. Together with the previous command, it demonstrates successful outbound relay delivery in this lab.

Active Directory

Task 1: Install Samba AD-DC.

Firstly, change the hostname to dc.rs.local as a FQDN for the domain controller

```
sang@2280031:~/Desktop$ sudo nano /etc/hostname
```

The domain-controller setup begins by opening /etc/hostname. A stable fully qualified host name is important because Samba and Kerberos refer to the domain controller by name.

```
  GNU nano 6.2                          /etc/hostname
dc.rs.local
```

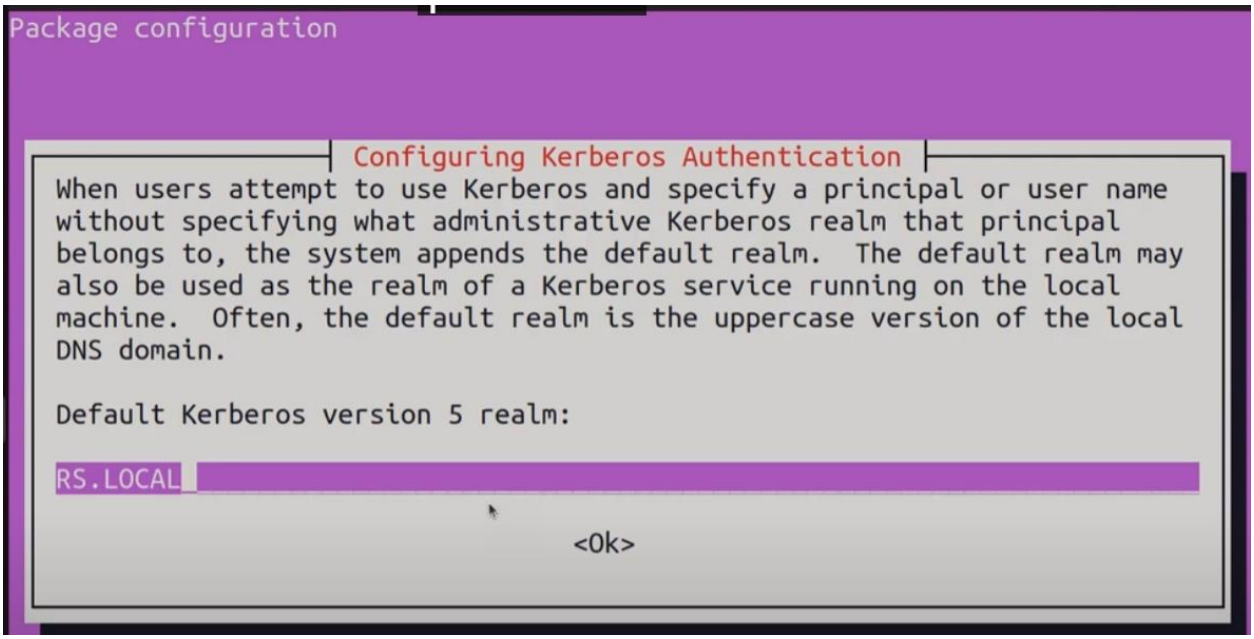
The hostname is set to dc.rs.local. This name identifies the Ubuntu machine that will provide the RS.LOCAL domain service.

Install samba, Kerberos 5, windows services and Linux services

```
sang@dc:~/Desktop$ sudo apt -y install samba krb5-config winbind smbclient
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  attr ibverbs-providers libboost-iostreams1.74.0 libboost-thread1.74.0
  libcephfs2 libgfapi0 libgfrpc0 libgfxdr0 libglusterfs0 libibverbs1 librados2
  librdmacm1 liburing2 python3-dnspython python3-gpg python3-markdown
  python3-pygments python3-requests-toolbelt python3-samba python3-tdb
  samba-common samba-common-bin samba-dsdb-modules samba-vfs-modules tdb-tools
Suggested packages:
  python3-sniffio python3-trio python-markdown-doc python-pygments-doc
  ttf-bitstream-vera bind9 bind9utils ctdb ldb-tools ntp | chrony
  smbldap-tools heimdal-clients cifs-utils libnss-winbind libpam-winbind
The following NEW packages will be installed:
  attr ibverbs-providers krb5-config libboost-iostreams1.74.0
  libboost-thread1.74.0 libcephfs2 libgfapi0 libgfrpc0 libgfxdr0 libglusterfs0
  libibverbs1 librados2 librdmacm1 liburing2 python3-dnspython python3-gpg
  python3-markdown python3-pygments python3-requests-toolbelt python3-samba
  python3-tdb samba samba-common samba-common-bin samba-dsdb-modules
  samba-vfs-modules smbclient tdb-tools winbind
```

The apt command installs Samba, Kerberos configuration, Winbind, and SMB client utilities. These packages supply the domain-controller and authentication components used in the following steps.

Configure the default Kerberos realm:



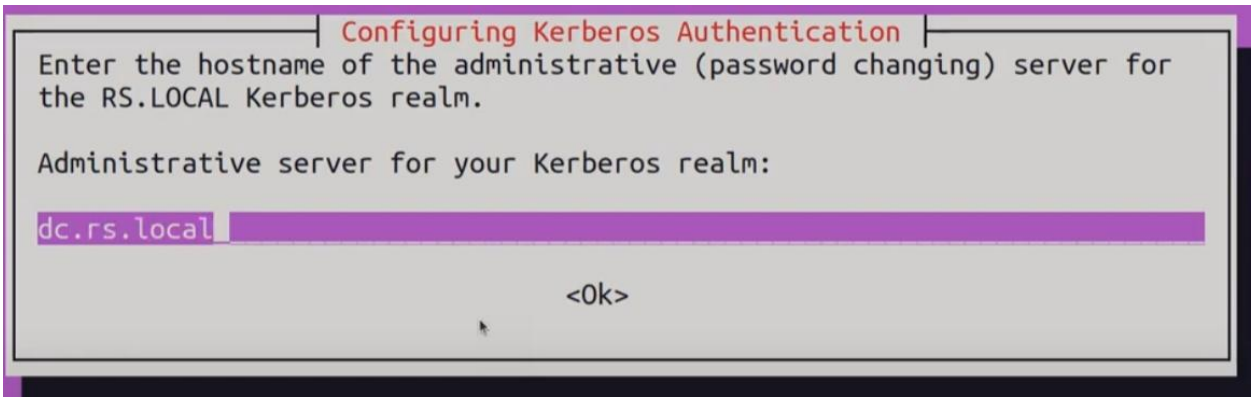
The default Kerberos realm is set to RS.LOCAL. Realm names are conventionally uppercase and must match the domain Samba will provision.

Configure Kerberos server for RS.LOCAL:



The Kerberos server is entered as dc.rs.local. Clients use that host name to locate the service that issues Kerberos tickets for the realm.

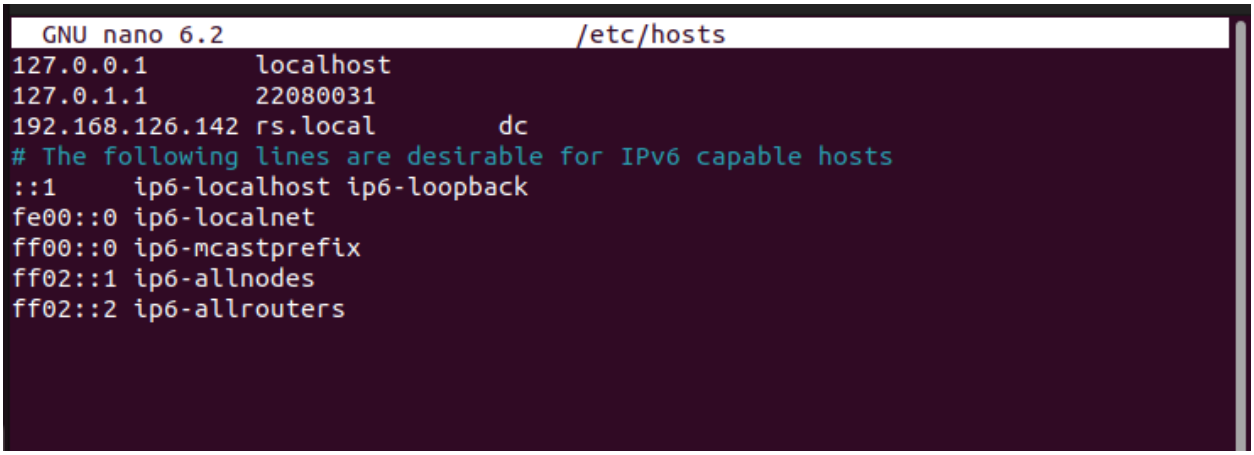
Configure the administrative server for your Kerberos realm:



The administrative Kerberos server is also set to dc.rs.local. Keeping both names consistent points realm administration to the same domain-controller VM.

Task 2: Configure Samba AD-DC

Configure the /etc/hosts file to resolve hostname (rs.local) to its IP address (192.168.126.142)



The hosts file associates the controller address 192.168.126.142 with its domain names. This provides local name resolution while the Samba DNS service is being prepared.

Backup the smb configuration by moving “smb.conf” to “smb.conf.install”

```
sang@dc:~/Desktop$ sudo mv /etc/samba/smb.conf /etc/samba/smb.conf.install
```

The existing smb.conf is moved to a backup name before provisioning. Samba can then generate a fresh domain-controller configuration without conflicting with the previous file.

Configure samba tool domain provision

```
sang@dc:/etc$ sudo samba-tool domain provision
Realm [RS.LOCAL]:
Domain [RS]:
Server Role (dc, member, standalone) [dc]:
DNS backend (SAMBA_INTERNAL, BIND9_FLATFILE, BIND9_DLZ, NONE) [SAMBA_INTERNAL]:

DNS forwarder IP address (write 'none' to disable forwarding) [192.168.126.142]:
  8.8.8.8,4.2.2.2
Administrator password:
Retype password:
INFO 2025-03-16 22:36:11,287 pid:2384 /usr/lib/python3/dist-packages/samba/provi
sion/__init__.py #2108: Looking up IPv4 addresses
INFO 2025-03-16 22:36:11,288 pid:2384 /usr/lib/python3/dist-packages/samba/provi
sion/__init__.py #2125: Looking up IPv6 addresses
WARNING 2025-03-16 22:36:11,288 pid:2384 /usr/lib/python3/dist-packages/samba/pr
ovision/__init__.py #2132: No IPv6 address will be assigned
INFO 2025-03-16 22:36:11,427 pid:2384 /usr/lib/python3/dist-packages/samba/provi
sion/__init__.py #2278: Setting up secrets.ldb
```

The samba-tool domain provision command creates the RS.LOCAL domain, administrator account, and internal DNS data. The choices shown use a domain-controller role and Samba's internal DNS backend.

Backup the Kerberos configuration files by copying “kerb5.conf” to “/etc/” folder.

```
sang@dc:~$ sudo cp /var/lib/samba/private/krb5.conf /etc/
```

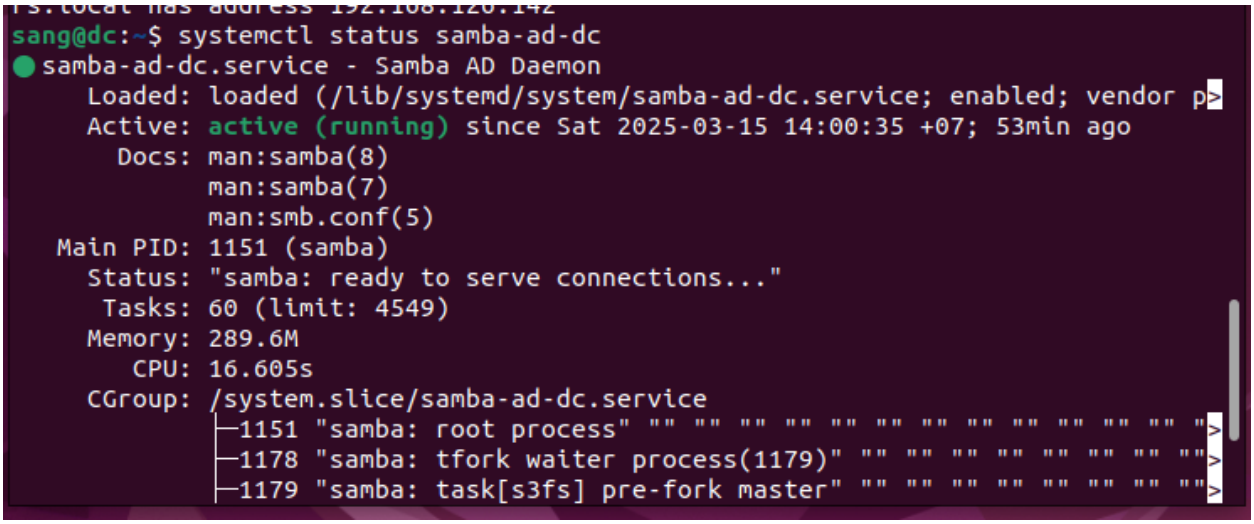
The generated Kerberos configuration is copied into /etc. This aligns local Kerberos tools with the realm settings produced by Samba provisioning.

Next, disable and unmask to remove any restrictions that prevent `samba-ad-dc` from being started. Then enable the service again:

```
sang@dc:~$ sudo systemctl disable --now smbd winbind
Synchronizing state of smbd.service with SysV service script with /lib/systemd/s
ystemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install disable smbd
Synchronizing state of winbind.service with SysV service script with /lib/system
d/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install disable winbind
Removed /etc/systemd/system/multi-user.target.wants/winbind.service.
Removed /etc/systemd/system/multi-user.target.wants/smbd.service.
sang@dc:~$ sudo systemctl unmask samba-ad-dc
Removed /etc/systemd/system/samba-ad-dc.service.
sang@dc:~$ sudo systemctl enable --now samba-ad-dc
Synchronizing state of samba-ad-dc.service with SysV service script with /lib/sy
stemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable samba-ad-dc
Created symlink /etc/systemd/system/multi-user.target.wants/samba-ad-dc.service
→ /lib/systemd/system/samba-ad-dc.service.
sang@dc:~$
```

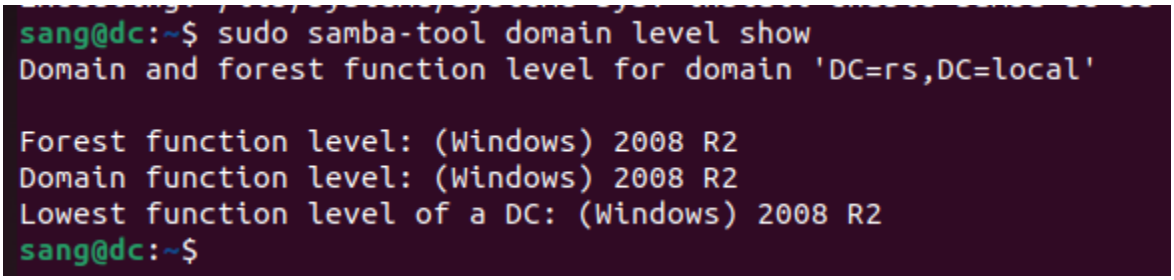
The standalone smbd and winbind services are disabled, and samba-ad-dc is unmasked and enabled. The domain-controller service manages the combined functions needed for this setup.

Check the samba AD DC status



The status output shows samba-ad-dc active and ready to serve connections. This confirms that the provisioned domain controller starts successfully.

Displays the functional levels of the AD domain and its forest.

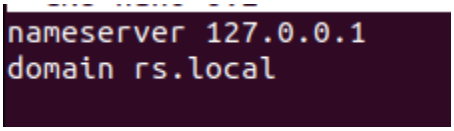


The domain-level command reports a Windows Server 2008 R2 functional level for the domain and forest. It is a compatibility setting reported by Samba, not evidence that a Windows Server is installed.

After that, remove the old DNS settings in the resolv.conf then make a new configuration:



The old resolver file is removed before the controller's DNS setting is changed. This lab step redirects local queries to the controller's own DNS service.



The replacement resolver entry uses 127.0.0.1 and the rs.local search domain. This lets the domain controller resolve its own domain records through its local DNS instance.

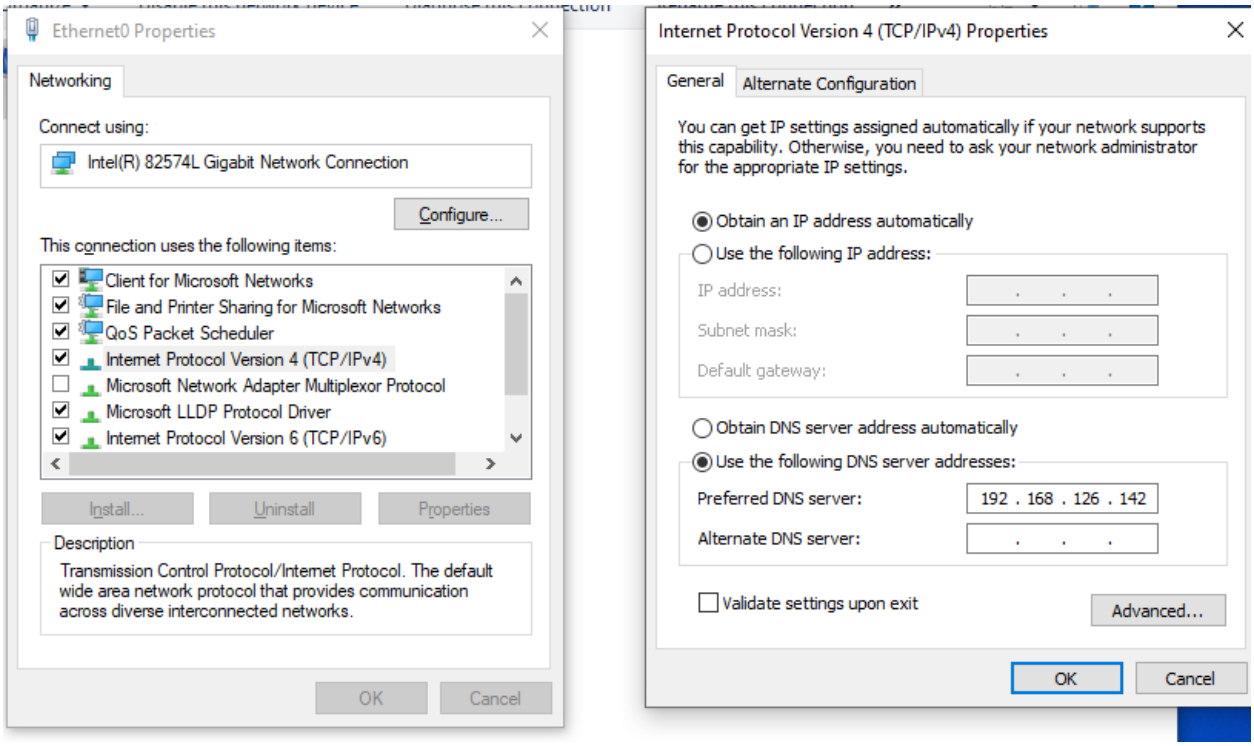
Task 3: Add client to the Domain Controller

Add user0:

```
sang@dc:~/Desktop$ sudo samba-tool user create user0
New Password:
Retype Password:
User 'user0' added successfully
sang@dc:~/Desktop$
```

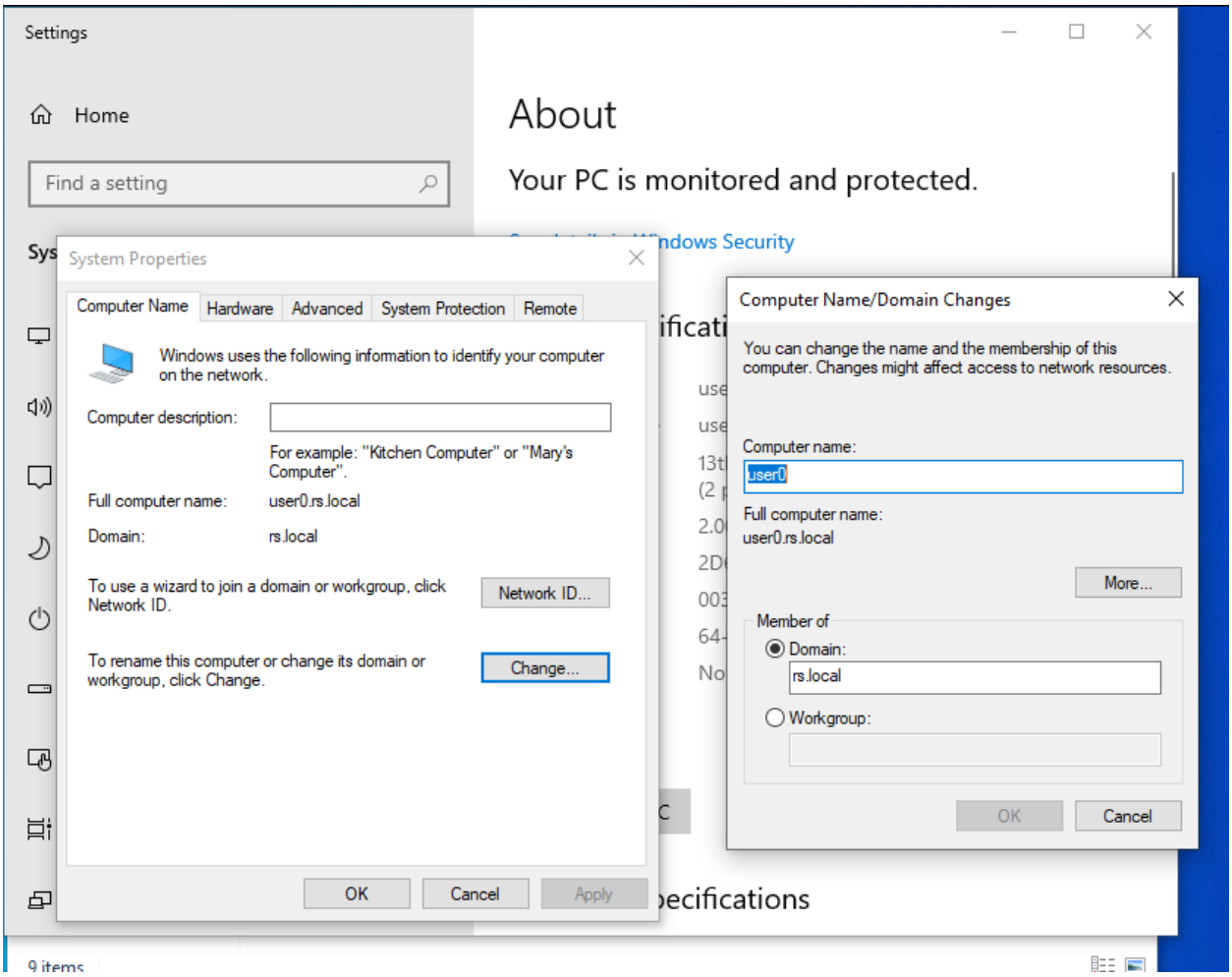
The samba-tool command creates a domain user named user0. The success message confirms that the account was added before the Windows client sign-in test.

After that, set the DNS to the IP address of the ubuntu samba machine in the window guest machine



The Windows client sets its preferred DNS server to 192.168.126.142. Active Directory discovery depends on DNS, so this points the client to the Samba controller's records.

Then enter the domain name:

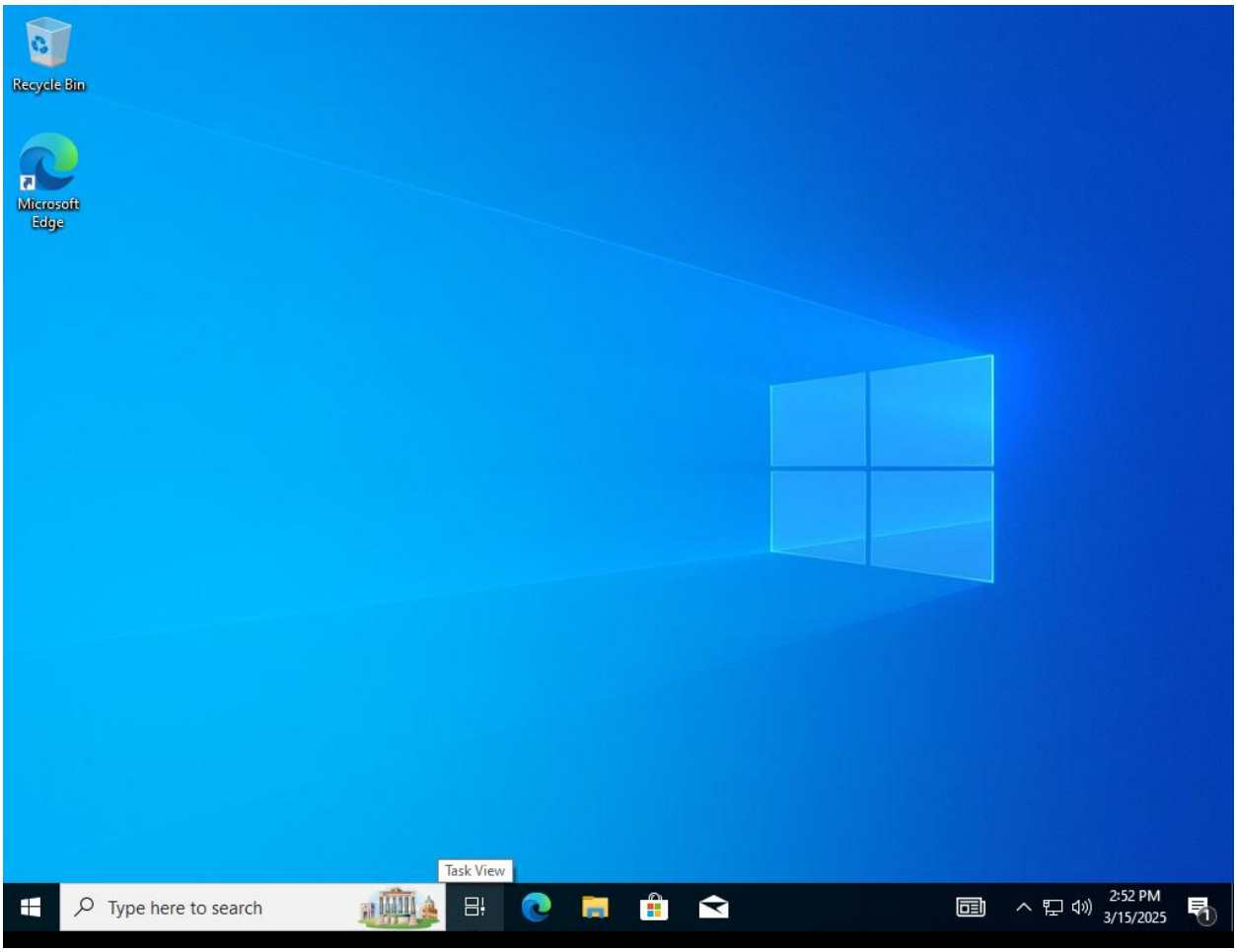


The Windows system-properties dialog enters rs.local as the domain to join. The join process uses the configured DNS server to find and contact the domain controller.

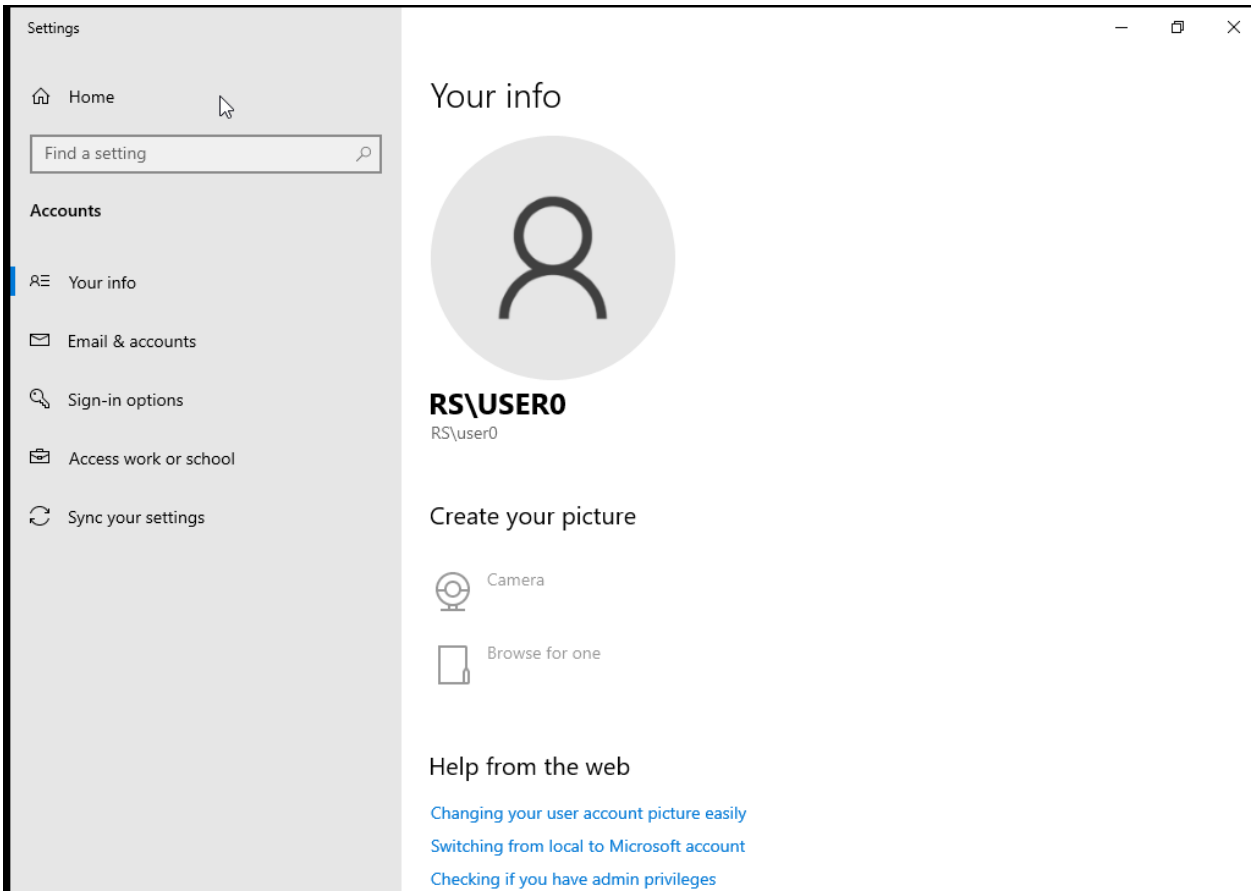
Finally, authenticate with user0 account:



The sign-in screen shows the domain account RS\user0. This tests authentication against the newly created domain identity rather than a local Windows account.

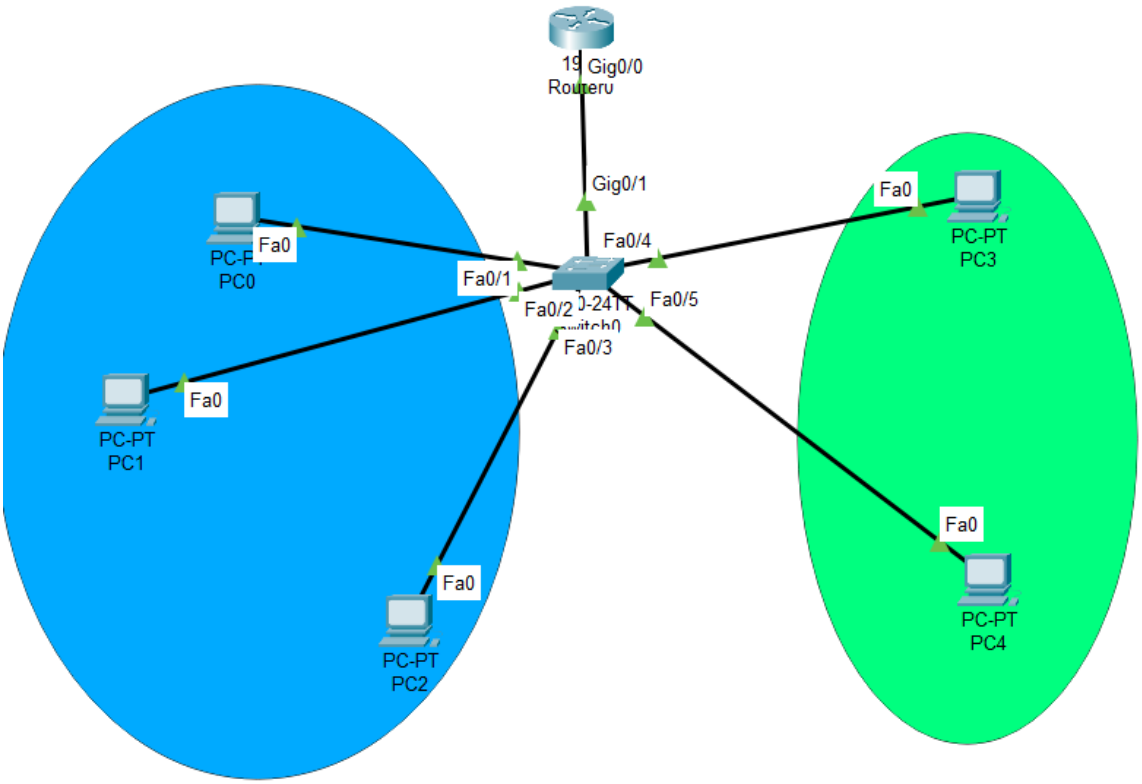


The Windows desktop appears after the domain sign-in. It indicates that the account completed a login on the joined client.



The account information panel identifies the signed-in user as RS\user0. This is a final visual check that the session is using the domain identity.

Configure a switch



The Packet Tracer diagram shows one switch connected to five PCs and a router. Three PCs are grouped for VLAN 2, two for VLAN 3, and the router link carries traffic between the VLANs.

Network table

Devices	Interface / sub-interface	VLAN	IP	Subnet mask
Switch 0	VLAN01	VLAN from 1 to 3	192.168.1.100	255.255.255.0
Router0	Gig0/0.1 (sub interface)	VLAN 1	192.168.1.1	255.255.255.0
	Gig0/0.2 (sub interface)	VLAN 2	192.168.2.1	255.255.255.0
	Gig0/0.3 (sub interface)	VLAN 3	192.168.3.1	255.255.255.0
PC 0	NIC	VLAN 2	192.168.2.2	255.255.255.0
PC 1	NIC	VLAN 2	192.168.2.3	255.255.255.0
PC 2	NIC	VLAN 2	192.168.2.4	255.255.255.0
PC 3	NIC	VLAN 3	192.168.3.2	255.255.255.0
PC 4	NIC	VLAN 3	192.168.3.3	255.255.255.0

The addressing table lists the switch management address, router subinterfaces, and PC addresses. It establishes the VLAN-to-subnet mapping used for the later switch and routing commands.

Task 1: Change hostname

```
Switch>enable
Switch#conf
Switch#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#host
Switch(config)#hostname
Switch(config)#hostname 22080031
22080031(config)#
```

The CLI command changes the switch host name to the lab identifier. A distinct prompt helps confirm that later commands are applied to the intended switch.

Task 2: Configure an administration password

Enable secret will store the password as a hash text.

```
22080031(config)#enable secret cisco123
22080031(config)#
```

The enable secret command sets the privileged-exec password. The value shown is a lab credential and should be replaced before using the configuration on a real device.

Task 3: Assign IP address

Create Vlan interface to assign IP address for the switch

```
22080031(config)#interface vlan 1
22080031(config-if)#ip add
22080031(config-if)#ip address 192.168.1.100 255.255.255.0
22080031(config-if)#no shu
22080031(config-if)#no shutdown

22080031(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

The switch virtual interface for VLAN 1 is assigned 192.168.1.100/24 and brought up. This address is used for remote management, including the SSH check at the end.

Task 4: Set password for SSH and Console access

Set password for console line:

```
22080031(config)#line console 0
22080031(config-line)#pass
22080031(config-line)#password cisco
22080031(config-line)#login
22080031(config-line)#exti
```

The console line is given a password and login is enabled. This controls direct console access, although the displayed example password is suitable only for a lab.

Set up SSH connection:

```
22080031(config)#ip domain-name example.com
22080031(config)#cry
22080031(config)#crypto ke
22080031(config)#crypto key g
22080031(config)#crypto key generate r
22080031(config)#crypto key generate rsa
The name for the keys will be: 22080031.example.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

22080031(config)#lin
*Mar 1 0:6:28.546: %SSH-5-ENABLED: SSH 1.99 has been enabled
22080031(config)#line vt
22080031(config)#line vty 0 4
22080031(config-line)#tran
22080031(config-line)#transport in
22080031(config-line)#transport input ss
22080031(config-line)#transport input ssh
22080031(config-line)#log
22080031(config-line)#login local
22080031(config-line)#
```

The switch is given a domain name and a 1024-bit RSA key is generated to enable SSH. The VTY lines are then restricted to SSH and local login, preventing Telnet access through those lines.

```
22080031(config)#username admin privilege 15 password adminpass
22080031(config)#
```

The local admin account is created for VTY authentication. The screenshot uses a visible lab password, which should be changed and stored securely outside a demonstration environment.

SSH configuration description

Command	Description
ip domain-name example.com	Sets the domain name for the switch, required for generating SSH keys.
cryto key generate rsa	Generates an RSA key pair for SSH encryption.
line vty 0 4	Defines virtual terminal (VTY) lines 0 to 4, which are used for remote access (SSH or Telnet). The switch allows up to five simultaneous remote connections (0-4).
transport input ssh	Restricts remote access to SSH only, disabling Telnet.
login local	Enables local authentication using configured usernames and passwords.
username admin privilege 15 password adminpass	Creates a user named "admin" with privilege level 15 (highest) and a password.

The SSH configuration table explains how the domain name, RSA key, VTY transport, and local account work together. The key permits encrypted sessions, while login local checks the configured user database.

Task 5: Create access-list for the switch

Create access list 1 that only permits IP of PC 0 to SSH in.

```
22080031(config)#access-list 1 permit 192.168.2.2
22080031(config)#lin
22080031(config)#line vt
22080031(config)#line vty 0 4
22080031(config-line)#acc
22080031(config-line)#access
22080031(config-line)#access-class 1 in
22080031(config-line)#
```

The access list permits 192.168.2.2, the address assigned to PC0, and applies that list to the VTY lines. It limits which source address may start a management session to the switch.

Task 6: Assign default gateway

```
22080031(config)#ip default-gateway 192.168.1.1
```

The switch default gateway is set to 192.168.1.1, the router address in the management subnet. This is required for a Layer 2 switch to reply to management traffic from other subnets.

Task 7: Configure VLANs

create VLAN 2 and 3

```
22080031(config)#vlan 2
22080031(config-vlan)#name VLAN2
22080031(config-vlan)#vlan 3
22080031(config-vlan)#name VLAN3
22080031(config-vlan)#exit
22080031(config)#
```

VLANs 2 and 3 are created and named. These logical segments separate the two PC groups even though all five PCs connect to the same physical switch.

Assign interfaces connected to PC 0, 1, 2 to vlan 2

```
22080031(config)#interface range fastEthernet 0/1-3
22080031(config-if-range)#swi
22080031(config-if-range)#switchport mo
22080031(config-if-range)#switchport mode acc
22080031(config-if-range)#switchport mode access
22080031(config-if-range)#sw
22080031(config-if-range)#switchport acc
22080031(config-if-range)#switchport access vl
22080031(config-if-range)#switchport access vlan 2
22080031(config-if-range)#
```

FastEthernet 0/1 through 0/3 are set as access ports in VLAN 2. Frames from PC0, PC1, and PC2 are assigned to the 192.168.2.0/24 segment.

Assign interfaces connected to PC 3, 4 to vlan 3

```
22080031(config)#interface range fastEthernet 0/4-5
22080031(config-if-range)#sw
22080031(config-if-range)#switchport mo
22080031(config-if-range)#switchport mode ac
22080031(config-if-range)#switchport mode access
22080031(config-if-range)#sw
22080031(config-if-range)#switchport ac
22080031(config-if-range)#switchport access vl
22080031(config-if-range)#switchport access vlan 3
22080031(config-if-range)#
```

FastEthernet 0/4 and 0/5 are set as access ports in VLAN 3. Frames from PC3 and PC4 are assigned to the 192.168.3.0/24 segment.

Assign interface Gi0/1, which is connected to router, as trunk mode

```
22080031(config)#interface gigabitEthernet 0/1
22080031(config-if)#sw
22080031(config-if)#switchport mod
22080031(config-if)#switchport mode trunk
22080031(config-if)#
```

GigabitEthernet 0/1 is set to trunk mode for the router connection. The trunk carries tagged traffic for multiple VLANs over a single link.

Go to the router, turn on the interface that connects to the switch

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shu
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
```

The router's physical switch-facing interface is brought up with no shutdown. Its subinterfaces cannot pass traffic while this parent link is administratively down.

Create sub-interfaces for the 3 VLANs in the router and assign IP to them.

```
Router(config-if)#exit
Router(config)#interfa
Router(config)#interface gi
Router(config)#interface gigabitEthernet 0/0.1
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.1, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation do
Router(config-subif)#encapsulation dot1Q 1
Router(config-subif)#ip aa
Router(config-subif)#ip ad
Router(config-subif)#ip address 192.168.1.1 255.255.255.0
Router(config-subif)#interface gigabitEthernet 0/0.2
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.2, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation do
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip add
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.3
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.3, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation do
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip add
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#
```

The router creates 802.1Q subinterfaces for VLANs 1, 2, and 3 at 192.168.1.1, 192.168.2.1, and 192.168.3.1. Each address serves as the gateway for its VLAN.

Enable ip routing

```
Router(config)#ip routing
```

The router accepts the ip routing command. The important operational check is that the subinterfaces and trunk permit packets to move between the VLAN subnets.

Task 8: Disable unneeded ports

```
22080031(config)#interface range fastEthernet 0/6-24
22080031(config-if-range)#shu
22080031(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down
```

Unused FastEthernet switch ports are shut down. Disabling unconnected access ports reduces opportunities for an unauthorized device to join the network.

```
22080031(config)#interface gigabitEthernet 0/2
22080031(config-if)#shu
22080031(config-if)#shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down
22080031(config-if)#
```

The unused GigabitEthernet 0/2 port is also shut down. The active router trunk remains on GigabitEthernet 0/1.

Save the configuration

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

The running configuration is copied to startup configuration. Saving prevents the switch or router settings from disappearing after a restart.

Verification:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=4ms TTL=128
Reply from 192.168.2.3: bytes=32 time=2ms TTL=128
Reply from 192.168.2.3: bytes=32 time=4ms TTL=128
Reply from 192.168.2.3: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\>
```

The first ping succeeds between two hosts in VLAN 2. This verifies access-port assignment and local Layer 2 connectivity inside that VLAN.

```
C:\>ping 192.168.3.2

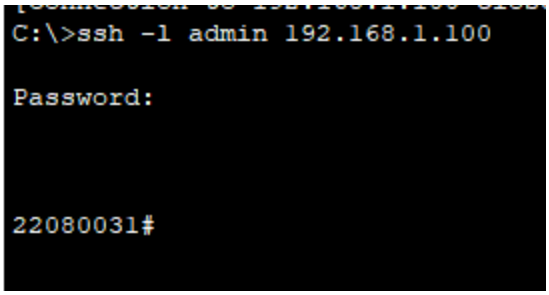
Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=2ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=3ms TTL=127

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

The second ping succeeds from VLAN 2 to a host in VLAN 3. This verifies that the trunk and router subinterfaces are forwarding traffic between subnets.



The SSH prompt shows a login to the switch management address 192.168.1.100. It demonstrates that the remote-management configuration accepts an authorized client connection.